Data bibliography

Orbital Technosphere: observer / Japan

01 Hajime Narukawa — AuthaGraph formulation
2022 formulation; native Cartofreako implementation
Projection and graticules are mathematical constructions. Horizontal periodicity supports the ST2 framing study.
https://doi.org/10.11212/jjca.60.1_1

02 Natural Earth — 1:10m physical vectors
Pinned local vectors used by the 29 October 2026 delivery; not new observations
Land, coastline, minor islands, lakes, rivers, ice and generalized bathymetry. Individual version files and component hashes are in the JSON ledger.
https://www.naturalearthdata.com/downloads/10m-physical-vectors/
Attribution / use: Public domain

03 CelesTrak — general perturbations orbital element groups
16 CSV groups retrieved 29 October 2026; SGP4 evaluated at 20:00 UTC that day
Active, Starlink, OneWeb, Kuiper, Qianfan, Hulianwang, GNSS, Iridium NEXT, Globalstar, ORBCOMM, resource, science, stations and three debris groups. Native seven-day age and six-hour future-epoch filters remain.
https://celestrak.org/NORAD/elements/
Attribution / use: CelesTrak source attribution and terms apply.

04 Natural Earth — Japan geographic boundary
1:10m boundary extracted for the 29 October 2026 delivery
Defines the Japan/Setouchi viewport. It does not turn national statistics into island-level measurements.
https://www.naturalearthdata.com/downloads/10m-cultural-vectors/10m-admin-0-countries/
Attribution / use: Public domain

Validation context only: NASA SSCWeb — retained validation reference
Historical 13 August 2026 reference; 29 October refresh timed out
Validation context only. It did not replace the newly retrieved CelesTrak orbital elements and is not a separate plotted live feed.
https://sscweb.gsfc.nasa.gov/WebServices/REST/

Measure and interpretation
SGP4 positions / 29 October 2026, 20:00 UTC
CelesTrak elements; seven-day age and six-hour future filters

Japan companion
Celestial directions calculated from Setouchi, 34.45°N 133.99°E, with a geographic locator; not a geographic crop or observed satellite photograph.

Source ledger
Pinned input files, complete source manifests and checksums:
https://bdekoz.github.io/cartofreako/docs/pages/gallery/setouchi-2028-bibliography.json

SGP4 positions / 29 October 2026, 20:00 UTC

CelesTrak elements; seven-day age and six-hour future filters

World dataset: 16 catalog groups retrieved 29 October 2026.

Debris / 52
Other Active / 362
Globalstar / 4
Iridium / 5
Navigation / 70
Earth Observation / 12
Science / 4
Starlink / 471
OneWeb / 41
Qianfan / 13
Hulianwang / 21
Kuiper / 23

Celestial directions, not a geographic Japan crop. The locator provides Setouchi context.

Setouchi observer (34.45°N, 133.99°E); the world observer uses its original reference.

The historical NASA SSC validation reference was retained after a refresh timeout.

Starlink names: 6 pt; sunbursts: 1.5×. Other names remain hidden.

Black x: debris. Intelsat: square. Remaining Earth observation: triangle.

Named groups follow memberships in the retained October source. Positions and observer are unchanged.

Yellow point: conceptual Setouchi reference (34.45°N, 133.99°E); venue unconfirmed.

Japan / geographic locator

Japan / Orbital technosphere Observer directions